Just Plane Awesome

A twisted rubber motor stores energy that spins a propeller. If the plane is light, strong, and trimmed to fly in a slow, gentle circle, it can stay up a long time.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 23	What Keeps a Plane Up?
Lesson 2	Fri Dec 4	Legal and Ready to Wind
Lesson 3	Fri Jan 22	Trim It, Then Test the Turns
Lesson 4	Fri Feb 19	Mock Tournament
Lab I1	Before Lesson 2	Forces, Parts, and Rules (home or an open Friday)
Lab I2	Before Lesson 3	Rubber Motor Energy Lab (home or an open Friday)
Lab I3	Before Lesson 4	Flight Log and Tool Kit Lab (home or an open Friday)

At the tournament

Pairs bring up to 2 student-built rubber-powered balsa planes, up to 4 motors, safety glasses, a tool kit, and an optional Flight Log (10% bonus), then get one 6-minute indoor flight period for test flights and up to 2 official flights; the longest official flight time wins.

Keep every handout for this event behind this divider, in order. Bring this binder to every Build & Test Fridays meeting.

Name _____

Just Plane Awesome: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can explain the event: 6-minute flight period, test flights vs official flights, up to 2 official flights, longest time wins.	L1 L4 I3	☐	☐
I can name the four forces on a plane (lift, weight, thrust, drag) and say what each does.	L1 I1	☐	☐
I can name the parts of my plane: wing, motor stick, stabilizer, rudder, propeller, prop hook, rear hook, rubber motor.	L1 L2 I1	☐	☐
I can build my plane myself at home, gluing balsa carefully and following the plans.	L1 L2	☐	☐
I can check my plane against the rules: wingspan 41.0 cm or less, mass 7.5 g or more without motor, one-piece plastic prop 15.5 cm or less, motor 2.0 g or less, labeled, made of wood, glue, and tissue.	L2 L4 I1	☐	☐
I can measure mass in grams with a scale and length in cm with a ruler.	L2 L3 I1	☐	☐
I can stretch-wind a rubber motor with a winder and count the turns.	L2 L3	☐	☐
I can explain how a twisted motor stores energy and changes it into motion.	L2 L3 I2	☐	☐
I can spot a stall, a dive, and a turn that is too tight, and name a fix for each.	L1 L3	☐	☐
I can make one small trim change at a time and test it.	L3 I2	☐	☐
I can time a flight with 3 timers and use the middle time.	L3 L4 I3	☐	☐
I can fill the official Flight Log with at least 10 flights: flight time, winder turns, and one more column (motor mass), all with units.	L2 L3 I2 I3	☐	☐
I can use my flight data to decide how many turns to wind.	L3 I2	☐	☐
I can pack a legal tool kit (winder, glue, tape, clay) with no extra rubber.	L4 I3	☐	☐
I can wear safety glasses the whole time and follow the rule that only students touch the plane.	L1 L2 L4	☐	☐

What Keeps a Plane Up?

TODAY I CAN

- I can explain the event: 6-minute flight period, test flights vs official flights, up to 2 official flights, longest time wins.
- I can name the four forces on a plane (lift, weight, thrust, drag) and say what each does.
- I can name the parts of my plane: wing, motor stick, stabilizer, rudder, propeller, prop hook, rear hook, rubber motor.
- I can build my plane myself at home, gluing balsa carefully and following the plans.
- I can spot a stall, a dive, and a turn that is too tight, and name a fix for each.
- I can wear safety glasses the whole time and follow the rule that only students touch the plane.

Welcome to Just Plane Awesome! You build a rubber-powered plane at home. At the tournament you get 6 minutes to fly it indoors: test flights, then up to 2 official flights. The longest official flight wins.

Today's steps: run it yourselves (22 minutes)

You need: 4 balsa toss gliders, modeling clay, 4 pairs of safety glasses, 2 stopwatches, 1 plane kit (and a finished demo plane if there is one), painter's tape. Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read the handout intro and 'The four forces' out loud. Tape a launch line on the floor.
2. Both 4 min. Set the timer for 4 minutes. Hold a glider. Take turns pointing to where lift, weight, and drag act on it. A glider has no thrust: it trades height for distance.
3. Both 8 min. Glasses on. Set the timer for 8 minutes. Get an adult in the room to say 'Clear!' before each launch. Launch gently and level from the line while your partner times it. Try no clay, a pea-size bit on the nose, then more, and fill 'Glider tests' (stall: nose up, then drops; dive: nose down fast).
4. Both 4 min. Set the timer for 4 minutes. Open the kit (or look at the demo plane). Find each part in 'Plane parts' and point to it for your partner. Do not wind the demo plane.
5. Both 3 min. Set the timer for 3 minutes. Read the 'At-home build log' and the at-home task. Then check your glider table with the answer card.

Check your work: Ask your Event Captain for the Lesson 1 answer card. Check that you named all four forces and that your glider table matches the card: too much nose clay dives, too little stalls.

The four forces

Lift	The upward push from air on the wing.
Weight	Gravity pulling the plane down.
Thrust	The forward pull from the spinning propeller.
Drag	Air pushing back on the plane and slowing it down.

Plane parts

wing **motor stick** **stabilizer (small back wing)** **rudder (tail fin)** **propeller** **prop hook** **rear hook** **rubber motor**

Glider tests

Try	Nose clay (none, a little, more)	Time (sec)	Stall, dive, or smooth?
1			
2			
3			

At-home build log (you build, a grown-up helps you stay safe)

Date	What I built	Minutes	Problem or question

At home before next time: Do Independent Lab I1 (Forces, Parts, and Rules) before Lesson 2. Also start your plane with a grown-up nearby: check every kit part against the parts list, read the plans, watch the balsa gluing video, and build the first steps (small dots of glue). Fill in the build log each time you work. Engineer's Logbook: build and trim a Straw Hoop Glider (p. 13) for the Glider Maker badge.

Legal and Ready to Wind

TODAY I CAN

- I can name the parts of my plane: wing, motor stick, stabilizer, rudder, propeller, prop hook, rear hook, rubber motor.
- I can build my plane myself at home, gluing balsa carefully and following the plans.
- I can check my plane against the rules: wingspan 41.0 cm or less, mass 7.5 g or more without motor, one-piece plastic prop 15.5 cm or less, motor 2.0 g or less, labeled, made of wood, glue, and tissue.
- I can measure mass in grams with a scale and length in cm with a ruler.
- I can stretch-wind a rubber motor with a winder and count the turns.
- I can explain how a twisted motor stores energy and changes it into motion.
- I can fill the official Flight Log with at least 10 flights: flight time, winder turns, and one more column (motor mass), all with units.
- I can wear safety glasses the whole time and follow the rule that only students touch the plane.

Before a plane can fly at the tournament, it must pass inspection. Then you need to wind it the right way.

Today's steps: run it yourselves (22 minutes)

You need: 0.01 g digital scale, meter stick and metric rulers, 2 winders, 4 rubber motors with O-rings, rubber lubricant, 4 pairs of safety glasses, 3 stopwatches, 2 official Flight Logs. Set the timer for each step.

1. Both 6 min. Set the timer for 6 minutes. Fill the 'Rule check' table: weigh the plane without its motor, measure the wingspan tip to tip, measure the propeller, and weigh each motor. Write units and mark OK or not.
2. Event Judge 3 min. Set the timer for 3 minutes. Read this out loud: Twisting the rubber stores energy. When it unwinds, it spins the propeller, which makes thrust. More turns store more energy, but too many can snap the motor.
3. Both 6 min. Glasses on. Set the timer for 6 minutes. Follow 'How to stretch-wind'. Partner A holds the plane; Partner B stretches the motor about 3 times its length and winds about 100 turns, counting out loud. Switch jobs.
4. Both 5 min. Set the timer for 5 minutes. Finished planes only. Get an adult in the room to say 'Clear!' before each launch. Hold the propeller, then release gently and level. Three people time; write the middle time, turns, and motor mass in 'Flight Log practice'.
5. Both 2 min. Set the timer for 2 minutes. Check your 'Rule check' with the answer card and write one fix to make at home.

Check your work: Ask your Event Captain for the Lesson 2 answer card. Compare your 'Rule check' numbers to the limits on the card and circle any rule your plane does not pass yet.

Rule check

Rule	Limit	Our plane	OK?
Mass without motor	7.5 g or more		
Wingspan	41.0 cm or less		
Propeller	One piece plastic, 15.5 cm or less		
Each motor (with O-rings)	2.0 g or less		
Made of	Wood, glue, tissue (no foam)		
Label	Team or school name on plane		

How to stretch-wind

1. Glasses on.
2. Partner holds the plane at the back of the motor stick.
3. Hook the winder to the front of the motor. Stretch the motor to about 3 times its length.
4. Wind and count. Walk slowly back toward the plane as you finish.
5. Hold the propeller, move the motor onto the prop hook, and launch gently and level.

Flight Log practice (copy to the official log later)

Flight #	Flight time (seconds)	Turns of winder	Motor mass (grams)	Notes
1				
2				
3				

At home before next time: Do Independent Lab I2 (Rubber Motor Energy Lab) before Lesson 3. Finish building your plane (you build, a grown-up keeps you safe) and write your school name on it. If you have a large indoor room, do 2 or 3 gentle glides with no winding over carpet with a grown-up watching, and add any flights to your Flight Log practice table. Engineer's Logbook: build a Rubber-Band Roller (p. 14) and test 10, 20, and 30 turns for the Energy Winder badge.

Trim It, Then Test the Turns

TODAY I CAN

- I can measure mass in grams with a scale and length in cm with a ruler.
- I can stretch-wind a rubber motor with a winder and count the turns.
- I can explain how a twisted motor stores energy and changes it into motion.
- I can spot a stall, a dive, and a turn that is too tight, and name a fix for each.
- I can make one small trim change at a time and test it.
- I can time a flight with 3 timers and use the middle time.
- I can fill the official Flight Log with at least 10 flights: flight time, winder turns, and one more column (motor mass), all with units.
- I can use my flight data to decide how many turns to wind.

Trim first so your plane flies smoothly. Then be a scientist: change only the number of turns and measure what happens.

Today's steps: run it yourselves (27 minutes)

You need: 0.01 g scale, 2 winders, rubber lubricant, modeling clay, 4 pairs of safety glasses, 3 stopwatches, clear tape and glue for repairs, painter's tape for the launch spot. Set the timer for each step.

1. Event Judge 3 min. Set the timer for 3 minutes. Read the 'Trim chart' out loud. Each partner names one problem and its fix without looking. Pick 3 timers.
2. Both 2 min. Set the timer for 2 minutes. Weigh your motor on the 0.01 g scale. Write its mass in grams in every row you fill today.
3. Both 8 min. Glasses on. Set the timer for 8 minutes. Get an adult in the room to say 'Clear!' before each launch. Wind about 200 turns, fly, name the problem with the 'Trim chart', make one small change, and fly again. Log each flight in 'Trim flights'.
4. Both 10 min. Glasses on. Set the timer for 10 minutes. Get an adult in the room to say 'Clear!' before each launch. Fly once at each of 3 or 4 turn counts (like 200, 300, 400), same plane and launch. Write the middle time in 'Turns test'. Stop if it hits the ceiling or the rubber bunches.
5. Both 4 min. Set the timer for 4 minutes. Plot 'Turns test' on the graph with even spacing. Answer 'Think about it', then check with the answer card.

Check your work: Ask your Event Captain for the Lesson 3 answer card. Compare each trim change you made to the fixes on the card, and compare your graph's pattern and answers to the card. Your numbers will be different, but the pattern should match.

Trim chart

Problem	What it looks like	Try this (one at a time)
Stall	Nose goes up, plane slows down, then drops	Add a tiny bit of clay to the nose, or slide the wing back a little
Dive	Nose points down and plane hits the floor fast	Remove some nose clay, or slide the wing forward a little
Turn too tight	Small circles, plane spirals down	Bend the rudder a little less
Flies straight into walls	No circle	Bend the rudder a little toward the side you want it to circle
Climbs too steeply at the start	Shoots up, may hit ceiling	Wind fewer turns until the trim is right

Trim flights

Flight #	Flight time (seconds)	Turns of winder	Motor mass (grams)	What I changed

Turns test

Flight #	Turns of winder	Flight time (seconds)	Motor mass (grams)	Hit ceiling? (yes/no)

Graph: turns (across the bottom) vs flight time in seconds (up the side)

Think about it

1. Why should you change only one thing at a time?

2. What happened to flight time as you added turns?

3. How many turns will you use at the tournament? Why?

At home before next time: Do Independent Lab I3 (Flight Log and Tool Kit Lab) before Lesson 4. Fix anything that broke today (you do the fixing). Bring your planes, motors, official Flight Log, and tool kit box to Lesson 4. Engineer's Logbook: do the Paper Plane Trim Lab (p. 15), fixing a stall, a dive, and a turn, for the Trim Tuner badge.

Mock Tournament

TODAY I CAN

- I can explain the event: 6-minute flight period, test flights vs official flights, up to 2 official flights, longest time wins.
- I can check my plane against the rules: wingspan 41.0 cm or less, mass 7.5 g or more without motor, one-piece plastic prop 15.5 cm or less, motor 2.0 g or less, labeled, made of wood, glue, and tissue.
- I can time a flight with 3 timers and use the middle time.
- I can pack a legal tool kit (winder, glue, tape, clay) with no extra rubber.
- I can wear safety glasses the whole time and follow the rule that only students touch the plane.

Mock tournament. Check in, fly your 6 minutes, and figure out your score just like the judges will.

Today's steps: run it yourselves (27 minutes)

You need: 0.01 g scale, meter stick, 3 stopwatches, safety glasses (use your own if you have them). Set the timer for each step.

1. Event Judge 7 min. Set the timer for 7 minutes. Check-in: read each row of 'Check-in' out loud. The pair measures and weighs its own planes, and the Judge marks Y or N.
2. Event Judge 14 min. Glasses on. Set the timer for 14 minutes. Give each pair a 6-minute flight period on a stopwatch. Get an adult in the room to say 'Clear!' before each launch. Three timers on each official flight: write all 3 in 'Official flights' and circle the middle one.
3. Both 3 min. Set the timer for 3 minutes. Fill 'Score': best official time, times 1.1 if the Flight Log passed. Check your math with the answer card.
4. Both 3 min. Set the timer for 3 minutes. Answer 'Before the tournament'.

Check your work: Ask your Event Captain for the Lesson 4 answer card and check your score math. Then look at every N in 'Check-in': each one must be fixed before the tournament.

Check-in

Rule	Plane 1 (Y/N)	Plane 2 (Y/N)
Wood, glue, and tissue (no foam)		
Labeled with team or school name		
Wingspan 41.0 cm or less		
Mass without motor 7.5 g or more		
One-piece plastic prop, 15.5 cm or less		
Each motor 2.0 g or less (up to 4)		
Tool kit has no extra rubber		
Flight Log: 10+ flights, time, turns, 1 more column		

Official flights

Flight	Timer 1 (sec)	Timer 2 (sec)	Timer 3 (sec)	Middle time (sec)
Official 1				
Official 2				

Score

Item	Result
Best official time (sec)	
Flight Log bonus? (x 1.1)	
Final score (sec)	
Tier (1 = no rule broken, 2 = rule broken)	

Before the tournament

1. What will you repair or change?

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2. What will you pack?

At home before next time: Do your repair list (you do the work). Add any last flights to the Flight Log. Pack for tournament day: up to 2 planes in a safe box, up to 4 motors, winder, tool kit (no extra rubber), Flight Log, and safety glasses. Engineer's Logbook: build a Hangar Box (p. 16) to carry your planes for the Hangar Builder badge, and pack with the Tournament-day checklist (p. 17).

Lab I1: Forces, Parts, and Rules

THIS LAB HELPS ME

- I can name the four forces on a plane (lift, weight, thrust, drag) and say what each does.
- I can name the parts of my plane: wing, motor stick, stabilizer, rudder, propeller, prop hook, rear hook, rubber motor.
- I can check my plane against the rules: wingspan 41.0 cm or less, mass 7.5 g or more without motor, one-piece plastic prop 15.5 cm or less, motor 2.0 g or less, labeled, made of wood, glue, and tissue.
- I can measure mass in grams with a scale and length in cm with a ruler.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Four forces act on every plane, and every plane must pass inspection. Do this lab alone or with your partner, at home or at the Open Lab table. Nothing gets thrown.

Steps (29 minutes)

You need: 3 scrap paper sheets and 1 paper strip, metric ruler or meter stick, 4 pennies, kit plans or plane picture card. Set the timer for each step.

1. Me 3 min. Set the timer for 3 minutes. Read 'The four forces' out loud. Point up, down, forward, and back as you say each one.
2. Me 5 min. Set the timer for 5 minutes. Drop test: hold a flat sheet and a crumpled paper ball at the same height and let go at the same time (drop, do not throw). Then try a sheet folded in half. Fill 'Drop test'.
3. Me 4 min. Set the timer for 4 minutes. Lift test: hold one end of the paper strip just under your bottom lip so it droops. Blow hard straight out over the top. Watch the strip. Read 'Why it lifts' and write what you saw.
4. Me 5 min. Set the timer for 5 minutes. With the ruler, put pencil marks on scrap paper 41.0 cm apart (the widest wingspan) and 15.5 cm apart (the widest propeller). Fill the length rows of 'Rules I can measure'.
5. Me 4 min. Set the timer for 4 minutes. Mass sense: a penny is about 2.5 g. Hold 3 pennies (7.5 g, the lightest plane allowed without its motor), then 1 penny (a motor must be less than this). Fill the mass rows.
6. Me 5 min. Set the timer for 5 minutes. Use your kit plans or the picture card. Draw your plane in 'Draw and label' and label all 8 parts from 'Plane parts'.
7. Me 3 min. Set the timer for 3 minutes. Answer 'Think about it', then check with the I1 answer card.

Check your work: Get the I1 answer card from the Open Lab bin (or ask your Event Captain at Lesson 2). Check 'Drop test', 'Rules I can measure', and 'Think about it'. Fix any you missed in a different color.

The four forces

Lift	The upward push from air on the wing.
Weight	Gravity pulling the plane down.
Thrust	The forward pull from the spinning propeller.
Drag	Air pushing back on the plane and slowing it down.

Drop test

Paper shape	Landed 1st, 2nd, or 3rd?	Why?
Flat sheet		
Crumpled ball		
Folded in half		

Why it lifts

Fast-moving air over the curved top of the strip pushes less than the still air under it, so the air below pushes the strip up. A plane's wing is shaped to use moving air to make lift, and it also turns air downward, so the air pushes the wing up.

Rules I can measure

Rule	Limit	What I found or felt
Wingspan	41.0 cm or less	
Propeller	One piece plastic, 15.5 cm or less	
Plane mass without motor	7.5 g or more (about 3 pennies)	
Each motor with O-rings	2.0 g or less (less than 1 penny)	
Made of	Wood, glue, tissue	
Label	Team or school name on the plane	

Plane parts

wing **motor stick** **stabilizer (small back wing)** **rudder (tail fin)** **propeller** **prop hook** **rear hook** **rubber motor**

Draw and label your plane (all 8 parts)

Think about it

1. Which force does the rubber motor make?

2. In the drop test, which paper landed last? Which force slowed it most?

3. A plane weighs 6.9 g without its motor. Is it legal? What could you add?

4. A plane's wingspan is 42.5 cm. Is it legal? Why or why not?

Bring to the next meeting: Bring this page and your build log to Lesson 2.

Lab I2: Rubber Motor Energy Lab

THIS LAB HELPS ME

- I can explain how a twisted motor stores energy and changes it into motion.
- I can make one small trim change at a time and test it.
- I can fill the official Flight Log with at least 10 flights: flight time, winder turns, and one more column (motor mass), all with units.
- I can use my flight data to decide how many turns to wind.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Your plane's motor is a twisted rubber band. In this lab you build a small rubber-band spinner and run the same test you will run with your plane: more twists, more time?

Steps (27 minutes)

You need: rubber band, pencil, paper clip, index card and tape, stopwatch, metric ruler. Set the timer for each step.

1. Me 3 min. Set the timer for 3 minutes. Read 'Energy words' out loud.
2. Me 5 min. Set the timer for 5 minutes. Follow 'Build the spinner'. Measure the hanging rubber band in cm and write it in the table.
3. Me 10 min. Set the timer for 10 minutes. Twist the card 10 times, let go, and time until the card first stops. Do it twice. Then do 20, 30, and 40 twists. Keep everything else the same. Fill 'Spinner log'. Stop at 40 twists.
4. Me 4 min. Set the timer for 4 minutes. Plot your spinner log on the graph: twists across the bottom, seconds up the side, with even spacing.
5. Me 3 min. Set the timer for 3 minutes. Answer 'Think about it'.
6. Me 2 min. Set the timer for 2 minutes. Check with the I2 answer card.

Check your work: Get the I2 answer card from the Open Lab bin (or ask your Event Captain at Lesson 3). Your times will be different, but your pattern should match the card. Fix any answers you missed in a different color.

Energy words

Stored energy	Energy waiting to be used. A twisted rubber band stores energy.
Energy of motion	Energy something has when it moves. The unwinding rubber spins your card (or propeller).
Heat	Some energy always turns into a little heat. Stretch a rubber band fast and touch it to your lip: it feels warm.
Fair test	Change only one thing. Keep everything else the same.

Build the spinner

1. Loop the rubber band over the middle of the pencil.
2. Lay the pencil across the gap between two stacks of books so the rubber band hangs down.
3. Hook the paper clip through the bottom of the rubber band.
4. Tape the paper clip to the middle of the top edge of the index card. The card hangs like a flag.
5. Keep your face away from the rubber band while you twist.

Spinner log (like a Flight Log)

Try	Twists (turns)	Spin time (seconds)	Rubber band length (cm)
1	10		
2	10		
3	20		
4	20		
5	30		
6	30		
7	40		
8	40		

Graph: twists (across the bottom) vs spin time in seconds (up the side)

Think about it

1. What was the only thing you changed? Why keep everything else the same?

2. What happened to spin time as you added twists?

3. Where was the energy stored? What did it turn into?

4. Which column in your plane's Flight Log is like the 'Twists' column here?

Bring to the next meeting: Bring your spinner log and graph to Lesson 3.

Lab I3: Flight Log and Tool Kit Lab

THIS LAB HELPS ME

- I can explain the event: 6-minute flight period, test flights vs official flights, up to 2 official flights, longest time wins.
- I can time a flight with 3 timers and use the middle time.
- I can fill the official Flight Log with at least 10 flights: flight time, winder turns, and one more column (motor mass), all with units.
- I can pack a legal tool kit (winder, glue, tape, clay) with no extra rubber.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. A complete Flight Log adds 10 percent to your best time. This lab gets your log, your math, and your tool kit ready for the mock tournament. Nothing gets flown.

Steps (28 minutes)

You need: official Flight Log page, your Lesson 2 and 3 flight tables, calculator (optional). Set the timer for each step.

1. Me 8 min. Set the timer for 8 minutes. Copy every real flight from your Lesson 2 and 3 pages onto the official Flight Log. Put units in the headers: seconds, turns, grams. Write the total number of flights at the top. Then tick off 'Flight Log check'.
2. Me 4 min. Set the timer for 4 minutes. Fill 'Middle time practice'. The middle time is the one between the other two.
3. Me 4 min. Set the timer for 4 minutes. Answer 'Score math'.
4. Me 5 min. Set the timer for 5 minutes. Fill 'Tool kit sort'. At home, pack your real tool kit box with only the allowed items.
5. Both 4 min. Set the timer for 4 minutes. Write 'Our 6-minute flight plan'. If your partner is here, agree on it together.
6. Me 3 min. Set the timer for 3 minutes. Check with the I3 answer card.

Check your work: Get the I3 answer card from the Open Lab bin (or ask your Event Captain at Lesson 4). Check 'Middle time practice', 'Score math', and 'Tool kit sort'. If your log has fewer than 10 flights, circle the number you still need.

Flight Log check (all three needed for the bonus)

- [ ] Uses the official Flight Log page from the Rules Manual
- [ ] 10 or more test flights recorded
- [ ] Column 2 (flight time, seconds) filled in for every flight
- [ ] Column 3 (turns of winder) filled in for every flight
- [ ] At least one more column filled in: motor mass (grams), motor width (inches), or ceiling height (feet)
- [ ] Units written in the headers
- [ ] School name and our names at the top

Middle time practice

Timer 1 (sec)	Timer 2 (sec)	Timer 3 (sec)	Middle time (sec)
12.4	13.1	12.8	
20.5	19.9	20.2	
31.0	30.2	30.7	

Score math

1. Your best official flight is 40 seconds and your Flight Log passes. What is your score?

_____

2. Your best official flight is 35.5 seconds and your log passes. What is your score?

_____

3. Your log has only 8 flights. Do you get the 10 percent bonus?

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4. You launch and forget to say TEST. What happens?

Tool kit sort

Item	Allowed? (yes/no)
Winder	
Glue	
Clear tape	
Modeling clay	
Spare O-rings	
Fine sandpaper	
Extra rubber motor strip	

Our 6-minute flight plan

1. Glasses on before we walk in.
2. Test flight at our best turn count. Say TEST.
3. Fix trim if needed (one change).
4. Official flight 1. Say OFFICIAL.
5. Official flight 2 (a second plane or re-wound motor). Say OFFICIAL.

Bring to the next meeting: Bring your official Flight Log, packed tool kit box, planes, and motors to Lesson 4.